GRAPH THEORY

Midterm — Task 1

Travelling Salesman Problem

SPPG Meal Delivery Simulator

Road-network routing and exact optimisation using Held–Karp

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COURSE

Graph Theory | Midterm coding task

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1 - ALGORITHM IDEA

From city streets to a TSP

An SPPG kitchen must deliver meal boxes to all schools inside a selected radius, then return to its starting point. We seek the shortest complete tour. Since Paris includes one-way streets, the travel cost from school A to B may differ from B to A: this is an asymmetric TSP.

Local OSM extract → directed road graph → Dijkstra distance matrix → Held–Karp tour

Method: dynamic programming

Let D[i][j] be the shortest driving distance from stop i to stop j. Stop 0 is the SPPG, and the other m stops are schools. Held–Karp stores the minimum cost of a route that starts at 0, visits exactly a subset S of schools, and finishes at school j.

DP[S, j] = min (DP[S \ {j}, i] + D[i][j]), i in S \ {j}

The program initialises single-school paths, builds larger subsets using bitmasks, adds the final return to the SPPG and reconstructs the optimal order using parent pointers.

Implementation

osm_data.py Reads a local Paris .osm file, extracts amenity=school objects, filters roads and uses NetworkX Dijkstra to create a directed distance matrix.

held_karp.py Computes the exact minimum-distance tour from this matrix and returns the route, distance and solver time.

app.py Provides an interactive Streamlit/Folium map to select the kitchen and radius, then shows the route and results.

The Paris demo uses a local OSM extract after issues with Overpass access; the original assignment specifies Surabaya.

02 - COMPLEXITY & RESULTS

Computational complexity

For m schools, Held–Karp runs in $O(m^2 \times 2^m)$ time and uses $O(m \times 2^m)$ memory. This is substantially better than trying all m! possible visit orders, but still grows exponentially. The current interface therefore limits exact solving to 15 schools.

The road-routing stage is separate: Dijkstra is executed from each selected stop on a graph with V road nodes and E road edges. With a heap, the overall shortest-path work is approximately $O((m + 1)(V + E) \log V)$.

Observed Paris demonstration

Metric	Observed value
Schools visited	11
Optimal tour distance	5.78 km
Held–Karp solver time	0.049141 s (about 49 ms)
Estimated travel time	11.6 min at 30 km/h
Full waiting time	More than 10 minutes

The 49 ms measures only Held–Karp after the distance matrix is ready. The long end-to-end delay also includes reading OSM XML, identifying schools, building the road graph and calculating shortest paths. The exact bottleneck has not yet been measured separately.

Limitations and next step

- The current demonstration uses Paris rather than Surabaya; this change needs acceptance for the assignment.
- Driving-access filters and the fixed 30 km/h speed are simplifications; traffic, stops and some turn restrictions are not represented.
- The local extract has limited coverage, and the 15-school cap prevents exact solving of larger cases.

Conclusion. The method produces an exact tour for the road-distance matrix: 11 schools were connected in a 5.78 km loop. The most useful next improvement is to cache a prepared road graph and school list, rather than reprocessing the XML on every run.